

LINUX LOG ANALYSIS REPORT

Suspicious Login Detection & Brute Force Investigation

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Date
June 2025
Environment
Simulated Kali Linux Lab — Ubuntu Server 22.04 VM
Log File Analysed
/var/log/auth.log
Tools Used
Linux Terminal — grep, awk, tail, sort, uniq, wc, journalctl
Scenario Type
Simulated Security Incident — Brute Force + Privilege Escalation
Report Version v1.0

NOTE: This report is based on a simulated lab environment created to demonstrate Linux log analysis skills as practised during SOC analyst training. The scenario, log entries, and findings reflect realistic attack patterns used in real-world cybersecurity investigations.

1. OBJECTIVE

The objective of this exercise was to simulate a real-world SOC investigation by analysing Linux authentication logs to detect suspicious activity. The specific goals were:

- Analyse /var/log/auth.log to identify failed and successful SSH login attempts
- Detect brute force attack patterns by identifying repeated login failures from the same IP address
- Investigate suspicious root access attempts and privilege escalation activity
- Identify an unknown user account that may have been created by an attacker
- Document findings in a professional SOC-style incident report
- Map observed attack behaviour to the MITRE ATT&CK framework

2. LAB ENVIRONMENT & SETUP

Parameter	Details
Operating System	Kali Linux (Attacker) + Ubuntu Server 22.04 LTS (Target VM)
Virtualisation	VirtualBox — both machines on same internal network

Log File Investigated	/var/log/auth.log
SSH Service	OpenSSH running on port 22
Log Analysis Tools	grep, awk, tail, head, cat, sort, uniq, wc -l, journalctl
Scenario Simulated	External brute force attack followed by successful root login
Total Log Lines	1,847 log entries across 6-hour capture window
Investigation Date	June 2025

3. INVESTIGATION — COMMANDS USED & OUTPUT

3.1 Viewing Recent Log Entries

The investigation began by viewing the most recent entries in the authentication log to understand baseline activity.

```
$ tail -n 50 /var/log/auth.log

Jun 14 02:14:33 ubuntu sshd[2341]: Failed password for root from 185.234.219.55 port 45231 ssh2
Jun 14 02:14:35 ubuntu sshd[2341]: Failed password for root from 185.234.219.55 port 45232 ssh2
Jun 14 02:14:37 ubuntu sshd[2341]: Failed password for admin from 185.234.219.55 port 45233 ssh2
Jun 14 02:14:39 ubuntu sshd[2341]: Failed password for ubuntu from 185.234.219.55 port 45234 ssh2
Jun 14 02:14:41 ubuntu sshd[2341]: Failed password for root from 185.234.219.55 port 45235 ssh2
Jun 14 02:17:02 ubuntu sshd[2398]: Accepted password for root from 185.234.219.55 port 45289 ssh2
Jun 14 02:17:03 ubuntu sshd[2399]: pam_unix(sshd:session): session opened for user root
```

FINDING: Even from just the last 50 lines, multiple rapid failed logins from the same IP (185.234.219.55) were immediately visible — followed by a successful root login. This is a textbook brute force attack pattern.

3.2 Counting Total Failed Login Attempts

The total number of failed login attempts was counted to understand the scale of the attack.

```
$ grep 'Failed password' /var/log/auth.log | wc -l

247
```

247 failed login attempts were recorded in the auth.log file. This is extremely abnormal — a legitimate user would have at most 3–5 failed attempts before resetting their password.

3.3 Identifying the Most Suspicious IP Addresses

The following command extracted all IPs from failed login entries, counted occurrences, and sorted from highest to lowest to identify the primary attacker.

```
$ grep 'Failed password' /var/log/auth.log | awk '{print $11}' | sort | uniq -c | sort -rn

    231 185.234.219.55
     12 192.168.1.104
      4 10.0.0.22
```

IP Address	Failed Attempts	Classification	Verdict
185.234.219.55	231 attempts	External / Public IP	 PRIMARY ATTACKER — Brute Force
192.168.1.104	12 attempts	Internal Network IP	 Suspicious — Investigate
10.0.0.22	4 attempts	Internal Network IP	 Normal — Likely mistyped password

3.4 Investigating the Primary Attacker IP

A deeper investigation was performed on the primary attacker IP 185.234.219.55 to understand the full scope of activity.

```
$ grep '185.234.219.55' /var/log/auth.log

Jun 14 02:11:04 ubuntu sshd[2301]: Failed password for root from 185.234.219.55 port 44891 ssh2
Jun 14 02:11:06 ubuntu sshd[2301]: Failed password for admin from 185.234.219.55 port 44892 ssh2
Jun 14 02:11:08 ubuntu sshd[2301]: Failed password for ubuntu from 185.234.219.55 port 44893 ssh2
Jun 14 02:11:10 ubuntu sshd[2301]: Failed password for test from 185.234.219.55 port 44894 ssh2
Jun 14 02:11:12 ubuntu sshd[2301]: Failed password for postgres from 185.234.219.55 port 44895 ssh2 ... [226 more failed attempts across various usernames] ...
Jun 14 02:17:02 ubuntu sshd[2398]: Accepted password for root from 185.234.219.55 port 45289 ssh2
Jun 14 02:17:03 ubuntu sshd[2399]: pam_unix(sshd:session): session opened for user root
Jun 14 02:18:45 ubuntu sshd[2401]: pam_unix(sshd:session): session opened for user backdoor
```

CRITICAL FINDING: The attacker from IP 185.234.219.55 successfully logged in as ROOT at 02:17:02 after 231 failed attempts — a successful brute force attack. The attacker then opened a second session as user 'backdoor' — a suspicious unknown account.

3.5 Checking for Root Login Attempts

```
$ grep 'root' /var/log/auth.log | grep 'Failed'

Jun 14 02:11:04 ubuntu sshd[2301]: Failed password for root from 185.234.219.55 port 44891 ssh2
Jun 14 02:11:06 ubuntu sshd[2301]: Failed password for root from 185.234.219.55 port 44892 ssh2
... [187 root login failures from 185.234.219.55] ...
```

```
$ grep 'Accepted' /var/log/auth.log

Jun 14 02:17:02 ubuntu sshd[2398]: Accepted password for root from 185.234.219.55
port 45289 ssh2
Jun 14 08:34:11 ubuntu sshd[3021]: Accepted password for subbalakshmi from
192.168.1.5 port 52341 ssh2
```

Login Type	Username	Source IP	Time	Verdict
Successful — SUSPICIOUS	root	185.234.219.55	02:17:02	🔴 Attacker gained root access
Successful — Normal	subbalakshmi	192.168.1.5	08:34:11	🟢 Legitimate user login

3.6 Discovering the Unknown 'backdoor' User Account

After the successful root login, the attacker created a new hidden user account. This was discovered by checking all system user accounts.

```
$ cat /etc/passwd | grep -v 'nologin\|false'

root:x:0:0:root:/root:/bin/bash
subbalakshmi:x:1000:1000:::/home/subbalakshmi:/bin/bash
backdoor:x:1001:1001:::/home/backdoor:/bin/bash $ last
| head -20

backdoor pts/1      185.234.219.55    Sun Jun 14 02:18   still logged in
root     pts/0      185.234.219.55    Sun Jun 14 02:17 - 02:18  (00:01)
subbalaksh pts/0    192.168.1.5      Sun Jun 14 08:34   still logged in
```

CRITICAL FINDING: An unknown user account named 'backdoor' was found on the system with UID 1001. This account was not created by any legitimate administrator. The attacker created this account after gaining root access to maintain persistent access to the system — even if the root password is later changed.

3.7 Attack Timeline Reconstruction

Time	Event	Source IP	Severity
02:11:04	First failed SSH login attempt — root account targeted	185.234.219.55	🟡 Medium
02:11:04–02:17:01	231 brute force attempts across multiple usernames	185.234.219.55	🔴 Critical
02:17:02	SUCCESSFUL root login — brute force attack succeeded	185.234.219.55	🔴 Critical
02:17:03	Root session opened — attacker has full system control	185.234.219.55	🔴 Critical
02:18:45	Unknown user 'backdoor' account created by attacker	185.234.219.55	🔴 Critical
02:18:45	Attacker logs in as 'backdoor' — persistent access established	185.234.219.55	🔴 Critical

08:34:11	Legitimate user 'subbalakshmi' logs in from internal IP	192.168.1.5	 Normal
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4. KEY FINDINGS SUMMARY

#	Finding	Severity	Evidence
1	Brute force attack — 231 failed logins from single IP in 6 minutes	 Critical	grep 'Failed' wc -l output = 247; 231 from one IP
2	Successful root login from external IP after brute force	 Critical	Accepted password for root from 185.234.219.55
3	Unknown backdoor user account created post-compromise	 Critical	cat /etc/passwd shows 'backdoor' user not created by admin
4	Attacker maintained persistent access via backdoor account	 Critical	last command shows backdoor user still logged in
5	12 failed logins from internal IP 192.168.1.104	 Medium	Could be insider threat or misconfigured service — investigate
6	Legitimate user login from known internal IP confirmed normal	 Low	subbalakshmi login from 192.168.1.5 — expected activity

5. MITRE ATT&CK MAPPING

The observed attack behaviour maps to the following MITRE ATT&CK techniques:

MITRE Technique	ID	Observed In This Investigation
Brute Force — Password Guessing	T1110.001	231 automated login attempts across multiple usernames
Valid Accounts — Root Account Abuse	T1078.003	Attacker used compromised root credentials to log in
Create Account — Local Account	T1136.001	Attacker created 'backdoor' user for persistent access
Persistence via Valid Account	T1078	Backdoor account used to maintain access after detection
Exploitation of Remote Services	T1210	SSH service targeted as entry point for the attack

6. INCIDENT RESPONSE ACTIONS TAKEN

6.1 Immediate Containment

- Blocked IP 185.234.219.55 at the firewall level to stop ongoing attack
- Terminated all active sessions from the attacker IP using `pkill -u backdoor`
- Disabled the backdoor user account immediately: `usermod -L backdoor`
- Isolated the compromised machine from the network pending full investigation

6.2 Eradication

- Deleted the backdoor user account: `userdel -r backdoor`
- Changed root password to a strong randomly generated credential
- Reviewed `/etc/sudoers` for any unauthorised privilege escalation entries
- Checked for any cron jobs or startup scripts added by the attacker

6.3 Recovery & Hardening

- Disabled direct root SSH login: set `PermitRootLogin no` in `/etc/ssh/sshd_config`
- Implemented fail2ban to automatically block IPs after 5 failed login attempts
- Enabled SSH key-based authentication and disabled password authentication
- Reviewed all user accounts and removed any other suspicious accounts

6.4 Lessons Learned

- Root SSH login should never be enabled on a production server
- Password authentication over SSH is dangerous — key-based authentication is mandatory
- Fail2ban or equivalent brute force protection must be installed on all SSH-exposed servers
- Regular log monitoring would have detected this attack within minutes instead of hours

7. SOC ANALYST RELEVANCE

Skill Practised	Real-World SOC Application
Reading <code>auth.log</code> with <code>grep</code> and <code>tail</code>	Tier 1 SOC analysts investigate SSH logs daily during incident response
Counting failed logins with <code>wc -l</code>	Quantifying attack scale helps SOC prioritise severity and escalation
Extracting & ranking IPs with <code>awk</code> + <code>sort</code>	Identifying top attacker IPs is the first step in any network-based investigation
Tracing attack timeline from logs	SOC analysts reconstruct attack timelines to understand breach scope
Identifying backdoor accounts in <code>/etc/passwd</code>	Checking for rogue accounts is a standard post-compromise investigation step
Mapping to MITRE ATT&CK	All SOC teams classify incidents using MITRE — essential for reporting and tooling
Writing structured IR report	Every SOC investigation ends with a documented report for escalation and audit

8. LESSONS LEARNED

Before this exercise, I understood that Linux log files existed but did not know how to read them or what to look for. Working through this simulated investigation taught me that a single log file — `auth.log` — contains a complete record of every login attempt on a Linux system, and that simple command line tools like `grep`, `awk`, and `sort` are enough to uncover serious security incidents within minutes.

The most important realisation was how obvious a brute force attack looks in the logs — 231 failed attempts from the same IP address in 6 minutes is impossible to miss once you know what to search for. I also learned that attackers do not stop after gaining access — they immediately try to create persistent backdoor accounts so they can return even if the original vulnerability is fixed.

This project gave me a clear understanding of why log monitoring is the foundation of SOC work, and why tools like Splunk exist — to automate exactly this kind of search across thousands of machines simultaneously rather than investigating one log file at a time manually.

9. TOOLS & REFERENCES

Tool / Resource	Purpose	Cost
Kali Linux	Attack simulation and security testing environment	Free
Ubuntu Server 22.04	Target machine running SSH service	Free
VirtualBox	Virtualisation platform for lab setup	Free
grep, awk, tail, sort	Linux command line log analysis tools	Built-in
MITRE ATT&CK Framework	Attack classification and technique mapping	Free — attack.mitre.org
Professor Messer	Security+ reference for attack type definitions	Free
TryHackMe	SOC Level 1 path for supplementary lab practice	Free tier

- MITRE ATT&CK — <https://attack.mitre.org>
- Linux auth.log documentation — <https://linux.die.net/man/8/sshd>
- fail2ban documentation — <https://www.fail2ban.org>
- OpenSSH hardening guide — https://www.ssh.com/academy/ssh/sshd_config